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Mathematical model

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In Section 2.2 you derived the differential equation for the fraction of people $X(t)$ on Ledom Island infected with the flu virus

$$\frac{dX}{dt} = k X(t) (1 - X(t)),$$

with the initial condition

$$X(0) = 0.02.$$

Unfortunately, the parameter k was at that time unknown. However, because of the text in the cave you can now find a value for k !

Exercise A

1/1 point (graded)

The sentence in Relue was:

Twenty days after the foreigners arrived, already 730 Htam have fallen ill to this strange sickness.

The new information can be rewritten as:

☐ $X(20) = 750$

☒ $X(20) = 0.75$ ✓

☐ $X(20) = 730$

☐ $X(20) = 0.73$

Submit

You have used 2 of 2 attempts

i Answers are displayed within the problem

In the next section you are going to find a value for k to complete the model.

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